

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

Nature-Inspired Computation

Module Leader: Dr Edward Keedwell

TWO HOURS

Answer question 1 and two out of the three other questions. Question 1 is worth 40 marks. Other questions are worth 30 marks each.

Candidates are advised to spend FORTY-FIVE minutes on question 1 and THIRTY-FIVE minutes on each of the other questions.

No electronic calculators of any sort are to be used during the course of this examination.

ECM3412 (2015)

Compulsory question

1. (a) Write pseudocode for *an evolutionary algorithm* (EA). In your answer, you should include feasible **parameter** settings for each of the parameters associated with the algorithm.

(8 marks)

- (b) Optimisation algorithms must traverse a fitness landscape, that will be different for each problem encountered.

- i. Explain what is meant by the fitness 'landscape'.
- ii. Describe a feature in the landscape that would prove difficult for search algorithms to traverse, and thus find the optimum solution.
- iii. Explain why a simple algorithm such as a hillclimber is unlikely to find a good solution in complex search landscapes.

(9 marks)

- (c) Collectives in nature naturally exhibit a behaviour known as *emergence*. Briefly explain what the term emergence refers to in this context.

(3 marks)

- (d) Explain the following terms that appear in the descriptions of swarm intelligence algorithms:

- i. Cognitive component.
- ii. *gbest*.
- iii. Pheromone.
- iv. Autocatalytic process.

(8 marks)

- (e) Multi-objective evolutionary algorithms optimise a pareto-front or pareto-shell of solutions.

- i. Describe what is meant by the pareto front/shell.
- ii. Describe two desirable properties of this front/shell.
- iii. Explain which operations in the algorithm *NSGA-II* are designed to produce the desirable properties described in (ii).

(6 marks)

- (f) Perceptrons are simple computational techniques inspired by neurons in the human brain. With the aid of a labelled diagram, explain how a perceptron:

- i. processes incoming signals and produces an output.

- ii. learns from those signals to produce different output in subsequent iterations (epochs).

(6 marks)

(Total 40 marks)

Please Turn Over

2. Ant colony optimisation can be used for a number of different applications.

- (a) Describe the main algorithm and, briefly, the experimental findings in the paper Di Caro G., Dorigo M. (1998) "Ant Colonies for Adaptive Routing in Packet-Switched Communications Networks" in the *Proceedings of PPSN - Lecture Notes in Computer Science* Vol. 1498, pp. 673-682.

(15 marks)

- (b) Ants in ant colony optimisation algorithms need a topology to traverse to solve an optimisation problem.

i. Explain what is meant by a construction graph in ant colony optimisation.

(5 marks)

ii. Draw a construction graph for a problem with 5 decision variables and 3 choices (decisions) per variable.

(5 marks)

iii. Describe the experiments you would conduct to test the ACO formulation above. In your answer pay particular attention to the parameter settings that you would test and the experimental conditions used.

(5 marks)

(Total 30 marks)

Please Turn Over

3. (a) Multi-Objective Evolutionary Algorithms differ in several ways from their single-objective counterparts.

i. Explain what is meant by the term *domination* in multi-objective evolutionary algorithms.

(5 marks)

ii. Describe a process of ranking solutions within a population using the domination principle and explain why this is necessary in multi-objective optimisation.

(5 marks)

iii. Explain why multi-objective optimisation is preferable to repeating a single-objective optimisation run with different objective weightings.

(5 marks)

(b) Each nature-inspired algorithm has a different method of varying the extent to which it *exploits* or *explores*. For each algorithm below, describe the parameters that are modified to achieve different exploitation and exploration capability and explain how varying these alters the behaviour of the algorithm.

i. Genetic Algorithms.

(5 marks)

ii. Particle Swarm Optimisation.

(5 marks)

iii. Ant Colony Optimisation.

(5 marks)

(Total 30 marks)

Please Turn Over

4. Imagine that you have been provided with a dataset for a local weather forecasting centre. This data consists of measurements for *temperature in degrees Celsius, barometric (atmospheric) pressure in millibars, rainfall in millimetres, wind velocity in kilometres per hour and wind direction (N,S,E,W)* in addition to a record of the predominant weather in the area each day (i.e. *bright and clear, cloudy, showers, heavy rainfall, storms*). The forecasting centre would like you to build a model that will predict the predominant weather for the local area for a given day given the hourly timeseries data for the previous 24 hours. They will provide you with five years of historical data for you to build your model with. You decide to do this using a neural network.

(a) Explain how you would represent the input and output data to the network.

(8 marks)

(b) Describe the neural network architecture that you would use to process the data and explain your decisions.

(8 marks)

(c) Explain how you would divide up the data and describe the strategy that you would use to prevent the network from overfitting.

(8 marks)

(d) Describe the runs you would conduct to determine the performance of the neural network on the problem.

(6 marks)

(Total 30 marks)

End of Paper